DETERMINANT TRANSITION FORMULA FOR THE LOCAL COHOMOLOGY COMPARISON MAPS UNDER A CHANGE OF GENERATORS

THE STACKS PROJECT AUTHORS

Editorial note. The source ends 'Some details omitted' inside the Koszul-differential verification; the omitted step is the term-by-term cancellation argument for the alternating sum, carried as an explicit obligation.

Editorial note. Inputs lemma-extended-alternating-represented and lemma-application-nice-K-injective are same-chapter NOT carded (cross-line-claimed labels) and are declared external.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 1442 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Extended alternating complexes represent local cohomology. lemma-extended-alternating-represented (same chapter, NOT carded, declared external). Identifies the three Cech complexes with i_*R H_Z(F) up to canonical maps representing the identity.
- Local comparison maps from generators. Remark remark-supported-map-c-equations (same chapter, structural). Defines the maps c_{f_1,...,f_c} attached to a regular sequence cutting out Z.
- Functoriality of Cech complexes under the subdivision. lemma-application-nice-K-injective (same chapter, NOT carded, declared external). Underlies the compatibility used when comparing the three complexes.

Source reference. Original equation 0G7J: sources/stacks-perfect/pinned-original-source.tex, source lines 1107–1107. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0G7M: sources/stacks-perfect/pinned-original-source.tex, source lines 1185–1185. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 0G7I: sources/stacks-perfect/pinned-original-source.tex, source lines 1093–1093. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof

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has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 0G7Q: sources/stacks-perfect/pinned-original-source.tex, source lines 1267–1267. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *With X, $f_1, \ldots, f_c \in \Gamma(X, \mathcal{O}_X)$, and $\mathcal{F}$ as in Remark 0G7I. Let $a_{ji} \in \Gamma(X, \mathcal{O}_X)$ for $1 \leq i, j \leq c$ and set $g_j = \sum_{i=1,\ldots,c} a_{ji} f_i$. Assume $g_1, \ldots, g_c$ scheme theoretically cut out Z. If $\mathcal{F}$ is quasi-coherent, then*

$$c_{f_1,\ldots,f_c} = \det(a_{ji}) c_{g_1,\ldots,g_c}$$

where $c_{f_1,\ldots,f_c}$ and $c_{g_1,\ldots,g_c}$ are as in Remark 0G7Q.

Proof. We will prove that $c_{f_1,\ldots,f_c}(s) = \det(a_{ij}) c_{g_1,\ldots,g_c}(s)$ as global sections of $\mathcal{H}_Z(\mathcal{F})$ for any $s \in \mathcal{F}(X)$. This is sufficient since we then obtain the same result for section over any open subscheme of X. To do this, for $1 \leq i_0 < \ldots < i_p \leq c$ and $1 \leq j_0 < \ldots < j_q \leq c$ we denote $U_{i_0 \ldots i_p} \subset X$, $V_{j_0 \ldots j_q} \subset X$, and $W_{i_0 \ldots i_p, j_0 \ldots j_q} \subset X$ the open subscheme where $f_{i_0} \ldots f_{i_p}$ is invertible, $g_{j_0} \ldots g_{j_q}$ is invertible, and where $f_{i_0} \ldots f_{i_p} g_{j_0} \ldots g_{j_q}$ is invertible. We denote $\mathcal{F}_{i_0 \ldots i_p}$, resp. $\mathcal{F}'_{j_0 \ldots j_q}$ $\mathcal{F}''_{i_0 \ldots i_p, j_0 \ldots j_q}$ the pushforward to X of the restriction of $\mathcal{F}$ to $U_{i_0 \ldots i_p}$, resp. $V_{j_0 \ldots j_q}$, resp. $W_{i_0 \ldots i_p, j_0 \ldots j_q}$. Then we obtain three extended alternating Čech complexes

$$\mathcal{F}^\bullet : \mathcal{F} \to \bigoplus\nolimits_{i_0} \mathcal{F}_{i_0} \to \bigoplus\nolimits_{i_0 < i_1} \mathcal{F}_{i_0 i_1} \to \ldots$$

and

$$(\mathcal{F}')^\bullet : \mathcal{F} \to \bigoplus\nolimits_{j_0} \mathcal{F}'_{j_0} \to \bigoplus\nolimits_{j_0 < j_1} \mathcal{F}'_{j_0 j_1} \to \ldots$$

and

$$(\mathcal{F}'')^\bullet : \mathcal{F} \to \bigoplus\nolimits_{i_0} \mathcal{F}_{i_0} \oplus \bigoplus\nolimits_{j_0} \mathcal{F}'_{j_0} \to \bigoplus\nolimits_{i_0 < i_1} \mathcal{F}_{i_0 i_1} \oplus \bigoplus\nolimits_{i_0, j_0} \mathcal{F}''_{i_0, j_0} \oplus \bigoplus\nolimits_{j_0 < j_1} \mathcal{F}'_{j_0 j_1} \to \ldots$$

whose differentials are those used in defining (0G7J). There are maps of complexes

$$(\mathcal{F}'')^\bullet \to \mathcal{F}^\bullet \quad \text{and} \quad (\mathcal{F}'')^\bullet \to (\mathcal{F}')^\bullet$$

given by the projection maps on the terms (and hence inducing the identity map in degree 0). Observe that by Lemma 0G7M each of these complexes represents $i_* R\mathcal{H}_Z(\mathcal{F})$ and these maps represent the identity on this object. Thus it suffices to find an element

$$\sigma \in H^c((\mathcal{F}'')^\bullet(X))$$

mapping to $c_{f_1,\ldots,f_c}(s)$ and $\det(a_{ji}) c_{g_1,\ldots,g_c}(s)$ by these two maps. It turns out we can explicitly give a cocycle for σ. Namely, we take

$$\sigma_{1\ldots c} = \frac{s}{f_1 \ldots f_c} \in \mathcal{F}_{1\ldots c}(X) \quad \text{and} \quad \sigma'_{1\ldots c} = \frac{\det(a_{ji}) s}{g_1 \ldots g_c} \in \mathcal{F}'_{1\ldots c}(X)$$

and we take

$$\sigma_{i_0 \ldots i_p, j_0 \ldots j_{c-p-2}} = \frac{\lambda(i_0 \ldots i_p, j_0 \ldots j_{c-p-2}) s}{f_{i_0} \ldots f_{i_p} g_{j_0} \ldots g_{j_{c-p-2}}} \in \mathcal{F}''_{i_0 \ldots i_p, j_0 \ldots j_{c-p-2}}(X)$$

where $\lambda(i_0 \ldots i_p, j_0 \ldots j_{c-p-2})$ is the coefficient of $e_1 \wedge \ldots \wedge e_c$ in the formal expression

$$e_{i_0} \wedge \ldots \wedge e_{i_p} \wedge (a_{j_0 1} e_1 + \ldots + a_{j_0 c} e_c) \wedge \ldots \wedge (a_{j_{c-p-2} 1} e_1 + \ldots + a_{j_{c-p-2} c} e_c)$$

To verify that σ is a cocycle, we have to show for $1 \leq i_0 < \ldots < i_p \leq c$ and $1 \leq j_0 < \ldots < j_{c-p-1} \leq c$ that we have

$$\begin{aligned} 0 = &\sum\nolimits_{a=0,\ldots,p} (-1)^a f_{i_a} \lambda(i_0 \ldots \hat{i}_a \ldots i_p, j_0 \ldots j_{c-p-1}) \\ &+ \sum\nolimits_{b=0,\ldots,c-p-1} (-1)^{p+b+1} g_{j_b} \lambda(i_0 \ldots i_p, j_0 \ldots \hat{j}_b \ldots j_{c-p-1}) \end{aligned}$$

The easiest way to see this is perhaps to argue that the formal expression

$$\xi = e_{i_0} \wedge \ldots \wedge e_{i_p} \wedge (a_{j_0 1} e_1 + \ldots + a_{j_0 c} e_c) \wedge \ldots \wedge (a_{j_{c-p-1} 1} e_1 + \ldots + a_{j_{c-p-1} c} e_c)$$

is 0 as it is an element of the $(c+1)$st wedge power of the free module on $e_1, \ldots, e_c$ and that the expression above is the image of ξ under the Koszul differential sending $e_i \to f_i$. Some details omitted. $\square$